

HOMWORK 3

ANTONY PEREZ

6.1. Prove, that if $A : V \rightarrow W$ is an isomorphism and $v_1, v_2, \dots, v_n$ is a basis in V , then $Av_1, Av_2, \dots, Av_n$ is a basis in W .

Proof. Let w be a vector in W . Define $A^{-1} : W \rightarrow V$. We know $\sum_{i=1}^n a_i v_i = 0$ due to linear independence. $\square$

6.2. Find all right inverses to the 1×2 matrix $A = [1, 1]$. Conclude that A is not left invertible.

6.3. Find all left inverses of the column $(1, 2, 3)^T$

Proof. $[1 \ 0 \ 0]$ $\square$

6.4. Is the column $(1, 2, 3)^T$ right invertible? Justify.

Proof. Yes, $[1 \ 0 \ 0]$ $\square$

6.5. Find two matrices A and B that AB is invertible, but A and B are not.

6.6. Suppose the product AB is invertible. Show that A is right invertible and B is left invertible.

Proof. Let $AB : V \rightarrow W$. Since AB is invertible, define $(AB)^{-1} : W \rightarrow V$. $(AB)^{-1} = B^{-1}A^{-1}$. We see that $(BA)^{-1}AB = I$, so $B^{-1}A^{-1}AB = I$. $\square$

6.7. Let A and AB be invertible. Prove that B is invertible.

6.8. Let A be $n \times n$ matrix. Prove that if $A^2 = 0$ then A is not invertible.

6.9. Suppose $AB = 0$ for some non-zero matrix B . Can A be invertible? Justify.

6.10. Write matrices of the linear transformations T_1 and T_2 in $\mathbb{F}^5$, defined as follows: T_1 interchanges the coordinates x_2 and x_4 of the vector x , and T_2 just adds to the coordinate x_2 a times the coordinate x_4 , and does not change other coordinates.

$$T_1 \begin{pmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \\ x_5 \end{pmatrix} = \begin{pmatrix} x_1 \\ x_4 \\ x_3 \\ x_2 \\ x_5 \end{pmatrix}, \quad T_2 \begin{pmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \\ x_5 \end{pmatrix} = \begin{pmatrix} x_1 \\ x_2 + ax_4 \\ x_3 \\ x_4 \\ x_5 \end{pmatrix};$$

Show that T_1 and T_2 are invertible transformations, and write the matrices of the inverses.

6.11. Find the matrix of rotation in $\mathbb{R}^3$ through the angle α around the vector $(1, 2, 3)^T$. We assume the rotation is counterclockwise to the origin.

6.12. Give examples of 2×2 matrices such that:

- a) $A + B$ is not invertible although both A and B are invertible;
- b) $A + B$ is invertible although both A and B are not invertible;
- c) All of A, B and $A + B$ are invertible

6.13. Let A be an invertible symmetric matrix. Is the inverse of A symmetric? Justify.